seeed studio

INDUSTRIAL PRODUCT DATASHEET

SO-101 3D Printed Parts - LeRobot Open Source 6-DOF arm

SO-101 3D Printed Parts for the LeRobot Open Source 6-DOF arm featuring refined parts, a Servo Adaptor Board for Seeed Studio XIAO, and native LeRobot support for imitation learning and teleoperation.

SKU	UPDATED	PRICE
114993668	2026-09-05 04:28:02	$29.90(Excl. VAT)

Package Option	Pre-assembled	Servo Motors	Servo Control Board	3D-Printed Body Parts	Wrist Camera Bracket	USB Camera	Teleoperation	Imitation Learning
 Servo Motor Kit	-	✓	✓	-	-	-	✓	-
 Unassembled Bundle	-	✓	✓	✓	✓	-	✓	-
 Assembled Arm	✓	✓	✓	✓	✓	✓	✓	✓

Note: Servo Motor Kit + 3D Printed Parts = Unassembled Bundle

Overview

Package Option	Pre-assembled	Servo Motors	Servo Control Board	3D-Printed Body Parts	Wrist Camera Bracket	USB Camera	Teleoperation	Imitation Learning
 Servo Motor Kit	-	✓	✓	-	-	-	✓	-
 Unassembled Bundle	-	✓	✓	✓	✓	-	✓	-
 Assembled Arm	✓	✓	✓	✓	✓	✓	✓	✓

Note: Servo Motor Kit + 3D Printed Parts = Unassembled Bundle

The SO-101 3D Printed Parts kit for the LeRobot Open Source 6-DOF arm includes refined parts for easier installation and a Servo Adaptor Board for Seeed Studio XIAO. It supports six STS3215 magnetic-encoder bus servos with 30 kg·cm torque, 12-bit absolute position feedback, and flexible kit options for imitation learning and teleoperation.

Key Features

Note:

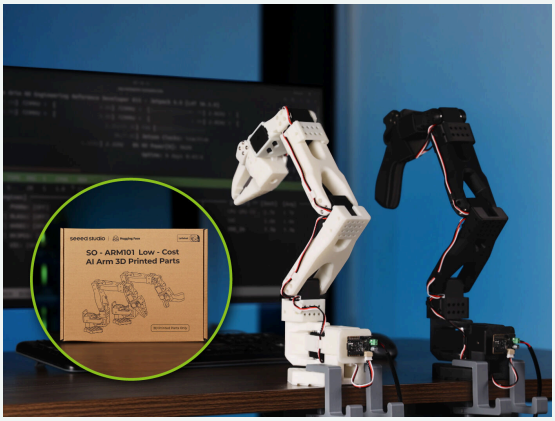
- We have finetune the 3D printed parts for easier installation and featuring Servo Adaptor Board for Seeed Studio XIAO. The parts will be slightly different than the official documents on Github.
- **6 Degree-of- Freedom(DOF) Arm** — Six STS3215 magnetic-encoder bus servos deliver 30 kg·cm torque per joint with 12-bit absolute position feedback for smooth, quiet, and repeatable motion
- **Hugging Face LeRobot Native Support** — Works out of the box with the LeRobot framework for imitation learning; collect teleoperation data and train ACT/Diffusion models with community-shared pipelines.
- **Affordable & Flexible Kit Options** — Low-cost 3D-printable robotic arm with flexible kit options. Select full-assembly, servo-only pack or unassembled parts to match your budget and project needs.
- **Fully Open-Source** — All hardware designs, BOM, firmware, and software are publicly available; modify, extend, and share freely

One-Stop Guided Setup & Cloud-Powered Robot AI Training

SenseCraft Robotics is an in-house developed, exclusive platform from Seeed Studio. It delivers an all-in-one robot-AI workflow featuring guided setup, cloud training and lightweight local validation. It lets you finish essential pre-training steps — device pairing, serial binding, camera binding, calibration check and teleoperation check — all in one place, greatly streamlining your setup for education labs, R&D PoCs and early-stage robot validation projects.

Product Gallery











Specification

Arm Type	SO-ARM101	
	Leader Arm	Follower Arm

Motor Configuration	1 × 7.4 V motor, 1:345 gear ratio — Joint 2	12x (12V) 1:345 gear ratio motors for all joints
	2 × 7.4 V motors, 1:191 gear ratio — Joints 1 & 3	
	3 × 7.4 V motors, 1:147 gear ratio — Joints 4 & 5 & Gripper	
Power Supply	DC 5V3A	DC 12V2A
DoF	5+1	
Payload	500g	
Angle sensor	12-bit magnetic encoder	
Operating Temperature Range	0℃～40℃	
Communication Method	UART	
Control Method	PC	

	SO-101 Servo Motor Kit	SO-101 Unassembled Bundle	SO-ARM101 Pro Kit
Price	$189.00	$295.00	$299.00
Assembly	Structural parts to be 3D-printed by user	Assembly from cratch	Pre-assembled
USB Camera	Excluded	Excluded	Included
Best For	Users with existing 3D-printed frames or custom enclosures	Hobbyists who enjoy the build process and want a complete kit. Select accessories as you like.	Researchers, educators, and teams who want to skip the build and focus on AI
Time to First Experiment	8+ hours	2-3 hours	Under 30 minutes

Hardware Overview



Applications

- **Leader-Follower Teleoperation:** Collect robot demonstration data easily via leader-follower teleoperation, ready for imitation-learning model training.
- **Imitation Learning & Reinforcement Learning:** Teach your robot household-style skills like folding clothes with human demos. Gather data via leader-follower teleoperation, train imitation-learning policies on LeRobot, and watch your robot reproduce tricky real-world manipulation tasks. SO-101 works seamlessly with LeRobot for data collection, simulation, training, and deployment. Train and deploy custom grasp-and-place policies quickly on NVIDIA Jetson edge devices. Explore the tutorial: https://www.jetson-ai-lab.com/archive/lerobot.html
- **Community Inspiration:** Follow Nikodem Bartnik's hands-on SO-101 workflow: hardware assembly, servo control, custom policy training and real-time block-picking inference. Walk through key steps, fix typical issues and explore the open-source arm's potential. Our monthly project call is now open! Share your project demos for a chance to win prizes and discount coupons.
- **Quick Start Developer Resources:** Jumpstart your project with official tutorials, open-source repositories, and community support: Start building your first Physical AI Project in 30 mins with SenseCraft; Step-by-Step Guide: SO-ARM 101 Quick Start https://wiki.seeedstudio.com/lerobot_so100m_new/; Open-Source Code: Access PyTorch models, simulation tools, and datasets on the Hugging Face LeRobot GitHub Repository https://github.com/huggingface/lerobot?tab=readme-ov-file; Community & Support: Join discussions, share models, and get real-time help on the Seeed Discord Channel https://discord.gg/8TnwDdjFGU & FaceBook https://www.facebook.com/share/g/17rksDbg1k/. Join our Facebook Group to claim exclusive coupons worth up to $100 USD.

Part List

Leader Parts	12 pcs
Follower Parts	11 pcs
Clamps Parts	12 pcs
Flexible Gripper Parts	4 pcs

Compliance & Logistics

HSCODE	3926909090
USHSCODE	3926100000
UPC	
EUHSCODE	3926909705
COO	CHINA

Resource Links

- https://www.seeedstudio.com/SO-101-3D-printed-Enclosure-p-6428.html
- STS3215 Servo Motor Datasheet
- Finedtune 3D printed parts files
- Original SO-ARM101 3D printed parts
- Servo Adaptor Board for Seeed Studio XIAO Datasheet
- reComputer J301x Datasheet
- reComputer J401X Datasheet
- reComputer_mini_datasheet_V1.0

Seeed Technology Co.,Ltd.
9F, Building G3, International E City,
Zhongshanyuan Road, Nanshan, Shenzhen, China

Tel: +86 0755-80695676
Web: www.seeed.cc
Shop: www.seeedstudio.com